Machine Learning Development Lifecycle

Course: Capstone Project

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SECTION 1
FOUNDATIONS

What Is Machine Learning

- Systems that learn patterns from data
- Predictive, generative, decision-making
- Data-driven automation
- Core to AI

Why ML Needs a Lifecycle

- Systems evolve over time.
- Data affects performance.
- Need structure and reliability.
- Ensures reproducibility.

ML Project Challenges

- Data drift
- Experiment tracking
- Deployment friction
- Monitoring needs

ML vs. Traditional Software

- Deterministic vs. probabilistic behavior
- Code + data + model artifacts
- Continuous retraining
- Higher operational complexity

ML Lifecycle Overview

- Step1- Problem definition
- Step 2- Data pipeline
- Step 3- Modeling
- Step 4- Evaluation
- Step 5- Deployment
- Step 6- Monitoring
- Step 7- Continuous improvement

SECTION 2
PROBLEM DEFINITION & DATA

Step 1- Problem Definition

- Define Business objective
- Identify Success Metrics
- Understand Constraints
- Clarify Stakeholders

Step 2: Data Collection

- Data Sources: Logs, APIs, sensors
- Batch vs streaming
- Data Ownership
- Data Governance

Data Labeling Approaches

- Manual labeling
- Weak supervision
- Active learning
- Quality assurance

weak supervision
Heuristic rules (most common)
Use simple rules to label data:

Example: spam detection
IF email contains "free money" → label as spam

Step 3: Data Preprocessing

- Cleaning and Normalization
- Handling Missing values
- Feature Engineering
- Train/test split

Feature Stores

- Centralized feature definitions
- Reusable features
- Online vs. offline stores
- Reduces training-serving skew

Data Quality & Drift

- Schema validation
- Statistical checks
- Covariate drift
- Concept drift

In Class Activity 1

1. Visit https://www.kaggle.com/ to learn about the available dataset. Pick a dataset (choose one that allows you access or create an account). Then answer the following questions:
   - What type of data (structured, non-structured, or semi-structured)?
   - How many records?
   - Can this dataset be used to train an AI model? What would such an AI predict or be used for?
   - Would this data be sufficient for the use you just mentioned, or would you need more data? Please explain.

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SECTION 3
MODELING & EVALUATION

Step4: Model Development

- Model selection
- Baseline Models
- Hyperparameter Tuning
- Experiment tracking

Step5: Model Evaluation

- Classification
- Regression
- Ranking
- Calibration

Responsible AI Considerations

- Fairness
- Explainability
- Bias detection
- Privacy and security

SECTION 4
DEPLOYMENT & MLOPS

What Is MLOps

- DevOps + ML
- (merging the goals by Software Development team, IT team considering ML project)
- Automating ML lifecycle
- Collaboration between DS + engineering
- Ensures reliability

Why MLOps Matters

- Faster iteration
- Reproducibility
- Scalable deployment
- Continuous training

MLOps Architecture Overview

- Data pipeline
- Feature store
- Training pipeline
- Model registry
- Deployment pipeline
- Monitoring

Step 6: Model Deployment

- Batch inference
- Real-time inference
- Edge Deployment
- Latency considerations

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In Class Activity 2

Do a quick Research on

- Batch inference
- Real-time inference
- Edge Deployment
- Latency considerations

Step 7: Monitoring

- Data Drift
- Prediction Drift
- Latency
- Throughput

Model Governance

- Auditability
- Compliance
- Documentation
- Risk management

ML Testing Strategies

- Unit tests
- Data tests
- Model tests
- Integration tests

ML Lifecycle Summary

- Iterative process
- Data-centric
- Requires automation
- Requires monitoring

SECTION 5
CLOUD ML DEVELOPMENT ENVIRONMENTS

Cloud Platforms

- Google Vertex AI
- Azure ML
- AWS SageMaker
- TensorFlow

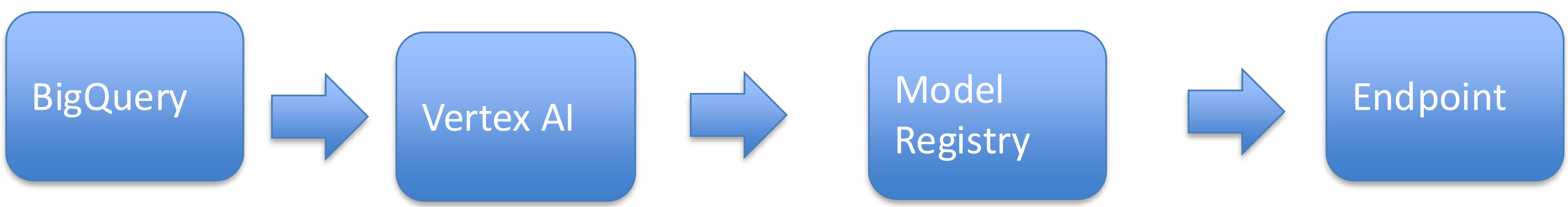
Google Cloud Lab: Overview

- Vertex AI Workbench
- Managed notebooks
- Integrated pipelines
- AutoML options

Google Cloud Lab: ML Workflow

- Data ingestion via BigQuery
- Training with Vertex AI
- Model registry
- Deployment endpoints

Google Cloud ML Architecture



Google Cloud Lab: Strengths

- Strong BigQuery integration
- Easy AutoML
- Scalable training
- Built-in experiment tracking

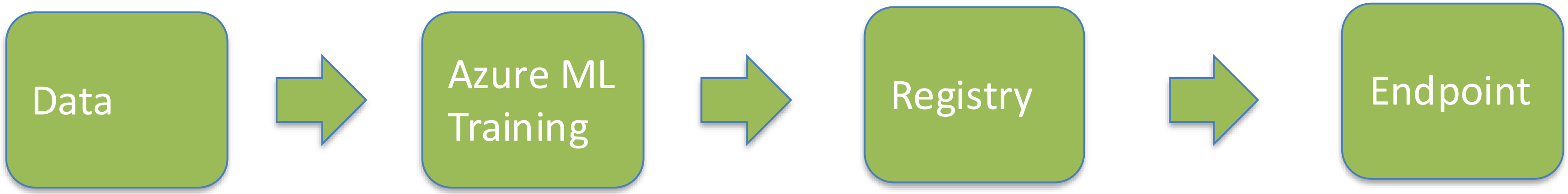
Microsoft Azure ML: Overview

- Azure ML Studio
- Designer pipelines
- Compute clusters
- Responsible AI dashboard

Azure ML: Workflow

- Data labeling
- Automated ML
- Training pipelines
- Managed endpoints

Azure ML Architecture



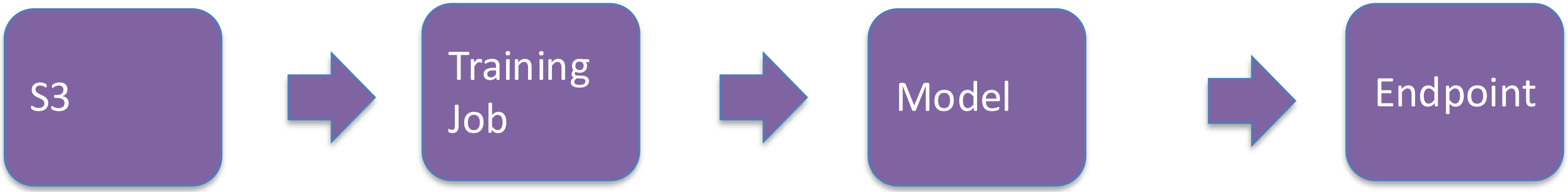
Azure ML: Strengths

- Enterprise governance
- Strong MLOps tooling
- Integrated monitoring
- Easy deployment

AWS SageMaker: Overview

- SageMaker Studio
- Built-in algorithms
- Training jobs
- Model hosting

AWS SageMaker Architecture



AWS SageMaker: Workflow

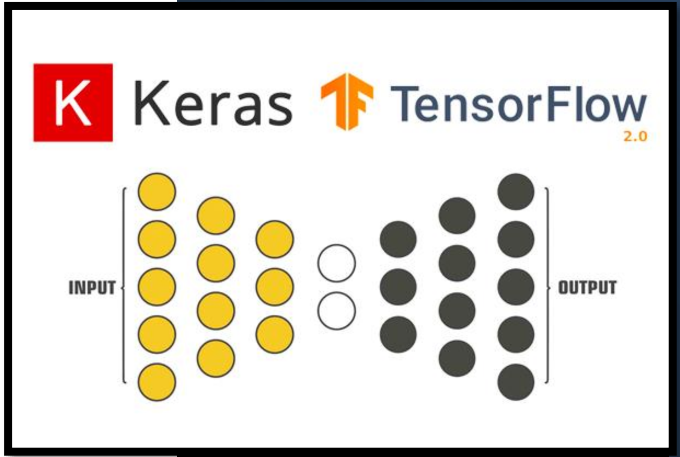
- Data prep with Data Wrangler
- Training jobs
- Model registry
- Real-time endpoints

AWS SageMaker: Strengths

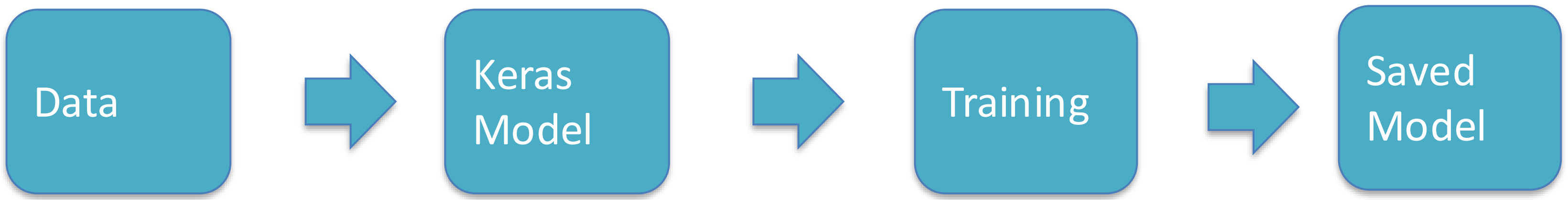
- Highly scalable
- Built-in algorithms
- Strong automation
- Cost-optimized training

TensorFlow + Python: Overview

- Open-source framework
- Keras API
- GPU acceleration
- Flexible modeling



TensorFlow Architecture



In Class Activity 3

- What Cloud Platform did you work on?
- What was the challenge(es) you had ?
- Let us know about your experiences.
- How long does it take for you to do your project?
- What mistakes you had ?
- What you learned about your mistake?

Highlights

- Lifecycle is iterative.
- Automation is essential.
- Monitoring is critical.

Final Thoughts

- ML is more than modeling.
- Data and operations matter.
- MLOps is the future of scalable AI.

YouTube Videos

- ***MLOps on Vertex AI***

https://docs.cloud.google.com/vertex-ai/docs/start/introduction-mlops

- ***MLOps By Andrew NG 1 hour***

https://www.youtube.com/watch?v=06-AZXmwHjo

- ***Machine Learning Lifecycle Overview***

https://www.youtube.com/watch?v=OMNHrfhdf0k

- ***TensorFlow in 10 Minutes***

https://www.youtube.com/watch?v=6_2hzRopPbQ

References

Machine Learning & MLOps (General)

- Microsoft: MLOps documentation https://learn.microsoft.com/azure/machine-learning/concept-model-management-and-deployment

Google Cloud (Vertex AI)

- Vertex AI documentation https://cloud.google.com/vertex-ai/docs

Microsoft Azure Machine Learning

- Azure ML documentation https://learn.microsoft.com/azure/machine-learning/

AWS SageMaker

- Amazon SageMaker documentation https://docs.aws.amazon.com/sagemaker/

TensorFlow & Python

- TensorFlow official documentation https://www.tensorflow.org/
- Keras API https://keras.io/
- TensorFlow tutorials https://www.tensorflow.org/tutorials

Responsible AI

- Microsoft Responsible AI https://www.microsoft.com/ai/responsible-ai